SPARKY'S SCIENCE EXPERIMENTS

Shadow Clock: Tell the Time with the Sun

Natural Science (Earth and space) · Ages 6+ · 10 minutes, then check every hour

Stand a stick in the ground and mark its shadow every hour to see how the sun seems to move across the sky.

Safety first

Sparky always says: Never look straight at the sun - it can hurt your eyes!

- Wear a hat and drink water when you are out in the sun.

You need

- A straight stick about 1 metre long (or a broom handle)
- Small stones or bottle caps
- Chalk or paper and a pencil
- A sunny, flat open space

Steps

1. On a sunny morning, stand the stick straight up in the ground or in a bucket of sand.
2. Put a stone at the tip of the shadow and write down the time.
3. Mark the shadow tip again every hour until the afternoon.
4. When was the shadow shortest? Which way did it point then?
5. Next sunny day, use your stones as a clock.

If it doesn't work, try these in order

1. Shadow hard to see? Wait for a sunny day - clouds make shadows fade.
2. Marks don't match the next day? Check the stick has not moved or tilted.

Why it works

The Earth spins once a day, so the sun seems to travel from east to west. Shadows point away from the sun: west in the morning, east in the afternoon. In almost all of South Africa the midday sun is in the northern sky, so the shortest shadow points south, and a shadow moves anticlockwise around the stick - the opposite way to clock hands! The shortest shadow is not exactly at 12:00: in Durban it is close to 12:00, in Cape Town closer to 13:00.

Did you know? Near the equator, in Kenya or Uganda, the midday sun is high overhead all year, so noon shadows are always short.

Ask Sparky: Will your clock still be right in winter? Why are shadows long in the morning?